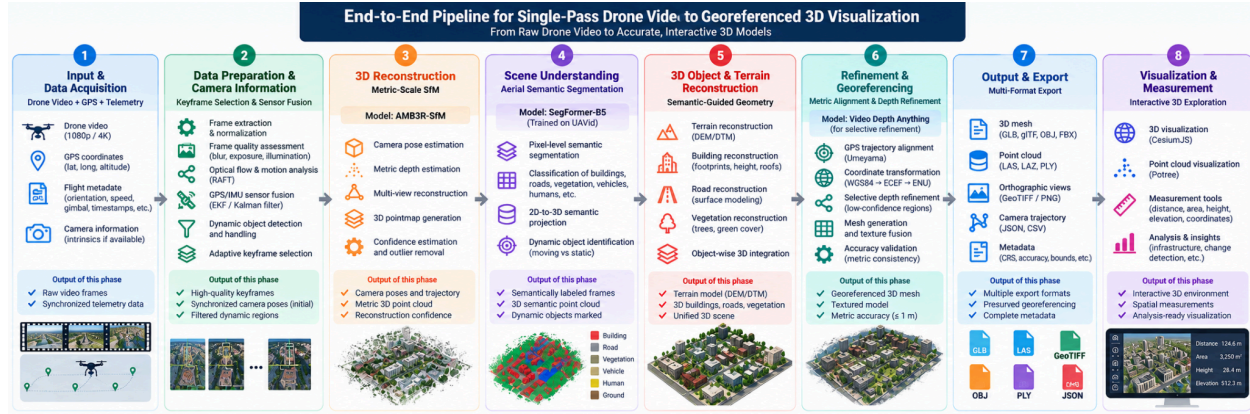


FINAL MODEL CONTEXT



Recommended ML / Technical Stack by Phase

Phase	Recommended Model / Technique	Main Purpose	Recommendation
1. Input	No ML model	Video + GPS + telemetry ingestion	Classical preprocessing
2. Data Preparation & Camera Information	RAFT + Kalman/EKF + adaptive keyframe selection	Motion estimation, camera-motion analysis, GPS fusion, frame selection	Recommended
3. 3D Reconstruction	AMB3R-SfM	Metric-scale camera poses, depth and 3D reconstruction	Primary model
4. Scene Understanding	SegFormer-B5 + UAVid	Aerial semantic segmentation	Primary candidate
5. Object & Terrain Reconstruction	Semantic-guided geometric reconstruction + Plane/Surface fitting	Terrain, roads, buildings, vegetation	Hybrid ML + geometry

6. Refinement & Georeferencing	Video Depth Anything + Umeyama + GPS/trajectory alignment	Depth refinement and metric georeferencing	Recommended
7. Output & Visualization	CesiumJS + Potree	3D visualization, point cloud, measurements	Recommended

Phase 1 → Input

Major technique

No ML model

Recommended technology

FFmpeg/OpenCV + GPS/telemetry parser

Why?

There is no reason to use an ML model here.

We simply need to reliably extract:

- video frames,
- timestamps,
- GPS coordinates,
- altitude,
- camera orientation,
- flight speed,
- gimbal information,
- camera parameters.

This should be deterministic preprocessing.

Final choice

Phase 1 → Video/GPS/Telemetry Ingestion

Phase 2 → Data Preparation & Camera Information

This is actually a combination of several techniques rather than one model.

Major model

RAFT

Purpose

RAFT provides dense optical flow between frames, which is useful for:

- camera-motion analysis,
- motion magnitude estimation,
- parallax estimation,
- scene-change detection,
- identifying unusual object motion,
- supporting dynamic-object detection,
- determining whether consecutive frames contain enough useful movement.

RAFT was designed for accurate optical flow and reported strong results on KITTI and Sintel; the original work reported KITTI F1-all of 5.10% and Sintel Final EPE of 2.855 px. ([arXiv](#))

GPS component

For GPS/IMU fusion, I would **not use an ML model**.

Use:

Extended Kalman Filter (EKF)

to combine:

- GPS,
- IMU,
- altitude,
- camera motion,
- flight telemetry.

Keyframe selection

Use a **rule/score-based adaptive keyframe selector**, using:

- GPS distance,
- RAFT motion,
- visual overlap,
- frame sharpness,
- scene changes,
- spatial baseline.

Final choice

Phase 2 → RAFT + EKF + Adaptive Keyframe Selection

This gives us a strong combination of learned motion estimation and deterministic sensor fusion.

Phase 3 → 3D Reconstruction

Major model: AMB3R-SfM

This is the most important model in the entire ML pipeline.

Purpose

AMB3R provides:

- camera poses,
- camera trajectory,
- metric depth,
- 3D pointmaps,
- multi-view reconstruction.

AMB3R was specifically designed for **metric-scale 3D reconstruction**, and its paper reports strong performance across camera pose, depth, metric-scale estimation and 3D reconstruction. It also provides large-scale SfM functionality without test-time optimization. ([arXiv](#))

Why AMB3R over the alternatives?

We previously considered:

- COLMAP
- VGGSfM
- DUS3R

- MAST3R
- AMB3R

For this project, AMB3R is the strongest fit because the requirement is not simply:

"Create a 3D model."

It is:

Create a metric-scale, georeferenceable 3D reconstruction from a single drone pass under computational constraints.

Therefore:

Final choice

Phase 3 → AMB3R-SfM

COLMAP remains the **fallback** for difficult reconstruction cases.

Phase 4 → Scene Understanding

Major model: SegFormer-B5

Dataset: UAVid

Here we should clearly distinguish:

UAVid = dataset

SegFormer = segmentation architecture

The segmentation model needs to classify aerial imagery into classes such as:

- building,
- road,
- vegetation,
- vehicle,
- human,
- ground/background.

SegFormer has direct evaluation on UAVid, and current UAVid benchmark listings include SegFormer results. ([Sota2](#))

A UAVID study reports SegFormer-B5 around **70.23% mIoU with test-time augmentation**, which makes it an attractive accuracy-first candidate. ([Sota2](#))

Why B5?

You explicitly said:

Accuracy is important; processing time comes second.

Therefore I would initially favor **B5 rather than B0/B1**.

B5 is heavier, but we don't necessarily need to run it on every raw video frame.

We can run segmentation on the **selected keyframes** and then propagate semantic information where appropriate.

Final choice

Phase 4 → SegFormer-B5 fine-tuned on UAVID + project-specific drone data

I would still benchmark B3 vs B5 before permanently locking the implementation, but **B5 is the accuracy-first starting point**.

Phase 5 → 3D Object & Terrain Reconstruction

This phase is different.

I would **not select one giant neural network**.

The best approach is:

Semantic-guided geometric reconstruction

Main techniques

Building → Plane/footprint/height reconstruction

Terrain → Ground extraction + surface fitting

Road → Semantic filtering + surface reconstruction

Vegetation → Semantic filtering + point-cloud reconstruction

The reason is important.

Once Phase 4 tells us:

"These points are buildings."

we can use geometric algorithms to determine:

- building footprint,
- roof planes,
- height,
- wall surfaces,
- terrain intersection.

For example, roof planes can be estimated using **RANSAC plane fitting**, while building footprints can be extracted from projected building points.

Why not another ML model?

Because geometric reconstruction has explicit physical structure.

A building wall is approximately planar.

A road is approximately a continuous surface.

Terrain is a continuous elevation field.

A geometric algorithm can exploit these constraints directly and gives more predictable measurements.

Final choice

**Phase 5 → Semantic-guided geometric reconstruction + RANSAC/plane fitting
+ surface reconstruction**

This is actually preferable to adding another large neural model.

Phase 6 → 3D Refinement & Georeferencing

This phase should use **two major technologies**, because they solve completely different problems.

6A → Depth Refinement

Major model: Video Depth Anything

Video Depth Anything is specifically designed for **temporally consistent depth estimation in long videos**. The paper reports state-of-the-art performance across multiple video benchmarks and emphasizes efficiency for long sequences. ([Open Access CVF](#))

It also provides multiple model sizes, including metric-depth versions. The official release reports the following A100 latency for 32-frame inputs:

- Small: **7.5 ms FP16**
- Large: **14 ms FP16**

under their benchmark setup. ([GitHub](#))

Important

We should **not run VDA over the entire video**.

Use it selectively:

AMB3R → confidence map → identify weak regions → VDA refinement → fuse improved depth

This preserves the near-real-time objective.

Recommended variant

For accuracy-first:

Video Depth Anything Large / Metric-Large

with smaller variants available if runtime becomes a problem.

6B → Georeferencing

Major technique: Umeyama Alignment

This should **not be an ML model**.

Use:

GPS trajectory + AMB3R camera trajectory → Umeyama similarity transformation

to estimate:

- scale,
- rotation,
- translation.

Then:

WGS84 → ECEF → ENU → aligned 3D reconstruction

Final choice

Phase 6 → Video Depth Anything + GPS/AMB3R Alignment + Umeyama

Phase 7 → Output & Visualization

Again, we don't need an ML model.

3D Visualization

CesiumJS

For:

- georeferenced GLB/glTF,
- terrain,
- buildings,
- roads,
- camera trajectory,
- measurements.

Point-cloud visualization

Potree

For:

- LAS/LAZ,
- large point clouds,
- point-cloud inspection,
- measurements,
- elevation profiles.

Final choice

Phase 7 → CesiumJS + Potree

Final Recommended Architecture

If you want the **one-line major technology/model for every phase**, I would put this in your architecture diagram:

Phase 1 → Video + GPS + Flight Telemetry

No ML model; deterministic ingestion.

Phase 2 → RAFT + EKF + Adaptive Keyframe Selection

Camera motion, sensor fusion, frame selection and dynamic-motion analysis.

Phase 3 → AMB3R-SfM

Metric-scale multi-view 3D reconstruction.

Phase 4 → SegFormer-B5 + UAVid

High-accuracy aerial semantic segmentation.

Phase 5 → Semantic-Guided Geometric Reconstruction

Terrain, roads, buildings and vegetation reconstruction.

Phase 6 → Video Depth Anything + Umeyama Alignment

Selective depth refinement + metric georeferencing.

Phase 7 → CesiumJS + Potree

Geospatial visualization, point-cloud visualization and measurement.

The most important models

If someone asks you **"What are the core AI models in your system?"**, don't list 10 models.

Say:

Our core AI stack consists of RAFT for motion analysis, AMB3R-SfM for metric 3D reconstruction, SegFormer trained on UAVid for aerial semantic segmentation, and Video Depth Anything for selective depth refinement. Georeferencing and final geometric reconstruction use established geometric methods such as EKF, Umeyama alignment, RANSAC and surface reconstruction.

That is a much more technically defensible architecture than trying to put an ML model into every single phase.